

Lucas Morais

Senior Backend Engineer • Java • Spring Boot • AWS • Microservices • Distributed Systems • Kubernetes • Docker • PostgreSQL • Kafka • CI/CD • ML Pipelines

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SUMMARY

Senior Backend Engineer with 9+ years designing and operating fault-tolerant, high-uptime systems for mission-critical industries. Specialist in Java/Spring Boot microservices, systems design, and data engineering pipelines — with proven delivery in the oil & gas sector where system failures carry direct operational and financial consequences. Author of CoDaF, an internal data persistence framework adopted organization-wide. Experienced across the full delivery cycle: architecture, implementation, analytics integration, code review, and engineering mentorship.

SKILLS

Languages: Java, Python, TypeScript, JavaScript, PHP, Rust

Frameworks: Spring Boot, Hibernate, React.js

Infrastructure: Docker, Kubernetes, Prometheus, Grafana, ELK Stack, CI/CD (GitLab CI, GitHub Actions), AWS EC2, Google Compute Engine, Azure VMs

Databases: PostgreSQL, SQL

Architecture: Microservices, Systems Design,DDD, TDD, REST APIs, Event-Driven Architecture, Clean Architecture

Data: Data Engineering, Real-Time Data Pipelines, Analytics, Scripting (Python), WITSML

Practices: Technical Leadership, Code Review, Engineering Mentorship, Agile

WORK EXPERIENCE

Senior Backend Engineer • Intelie

Feb 2020 – Present | Rio de Janeiro, Brazil • Remote

- Acted as interim technical lead during manager's leave, owning corporate stakeholder relationships and co-driving product roadmap prioritization alongside senior leadership — while sustaining full senior engineering responsibilities.
- Drove the client-reported ticket backlog from 24 to 3 open issues within weeks, restoring SLA health through the leadership transition.
- Serve as primary technical point of contact for a flagship enterprise client in the oil & gas sector, running recurring discovery sessions to translate operational requirements into engineering solutions.
- Re-architected server-side call orchestration, cutting active instances from 2,000+ to under 150 while preserving full functional coverage — eliminating server overload risk and infrastructure overhead.
- Led standardization of WITSML, an industry-standard oil & gas data-interchange protocol, aligning data ingestion with sector-wide interoperability requirements.
- Decomposed a legacy monolith into 6 independently deployable microservices using Java, Spring Boot, Docker, and Kubernetes, improving deployment agility and reducing cross-service dependencies across a platform supporting mission-critical oil & gas operations.
- Designed and shipped CoDaF, an internal data persistence framework adopted organization-wide: eliminated 3 redundant storage abstractions, enabled runtime schema provisioning, and reduced data inconsistency incidents across distributed services.
- Built real-time ML fault detection pipelines ingesting sensor telemetry from critical oil & gas infrastructure, enabling early anomaly detection before failures escalate, directly protecting operational uptime for 24/7 clients.

- Implemented OCR automation pipeline for ML data ingestion, reducing manual data preparation overhead for model training workflows.
- Led architecture and delivery of 3 platform initiatives as technical lead; defined engineering standards, conducted ongoing code reviews, and mentored junior engineers in DDD, TDD, and CI/CD best practices.
- Deployed and managed cloud infrastructure across AWS and Azure to support distributed service hosting and environment provisioning.
- Maintained 99.9%+ uptime SLAs across production systems serving clients operating continuous 24/7 infrastructure.

Stack: Java, Spring Boot, Hibernate, PostgreSQL, Docker, Kubernetes, Prometheus, AWS, Azure, REST APIs, Python, GitLab CI, GitHub Actions, WITSML

Software Engineer · Grupo Fratelli

Dec 2016 – Dec 2019 | Duque de Caxias, Brazil · On-site

- Architected and delivered a multi-warehouse management system end-to-end: inventory tracking, fiscal integration, and reporting across more than three distribution centers.
- Eliminated data silos between warehouse and finance operations by integrating SQL-backed inventory records with fiscal workflows, enabling real-time stock visibility across the organization.
- Developed backend business logic and server-side integrations handling fiscal workflows, inventory state, and reporting — delivering consistent feature velocity across a three-year engagement.

Stack: PHP, SQL, JavaScript

PROJECTS

CoDaF — Internal Data Persistence Framework | Intelie

Custom framework built to standardize data persistence across a distributed platform. Eliminated redundant storage abstractions, enabled runtime schema provisioning, and improved data consistency across microservices. Adopted organization-wide.

Real-Time Fault Detection Pipeline | Intelie

End-to-end ML data pipeline for oil & gas sensor telemetry: ingestion, processing, anomaly detection, and alerting. Built to operate within 99.9%+ uptime SLA requirements for 24/7 critical infrastructure clients.

Multi-Warehouse Management System | Grupo Fratelli

Full-stack inventory management, fiscal integration, and cross-site reporting system for more than three distribution centers. Delivered end-to-end over a three-year engagement.

EDUCATION

B.Sc. Computer Science — Universidade Estadual do Rio de Janeiro (UERJ)

Jul 2024 | Rio de Janeiro, Brazil

CERTIFICATIONS

- Java Multithreading, Concurrency & Performance Optimization — Udemy · Aug 2024
- Learn Rust by Building Real Applications — Udemy · Mar 2024
- CCNA — Cisco · Dec 2023
- Java Persistence API and Hibernate — Udemy · Jul 2022
- Unit Testing in Java: JUnit, Mockito & TDD — Udemy · Mar 2022
- Domain Driven Design (DDD) — Udemy · Jan 2022

LANGUAGES

English (Advanced) · Portuguese (Native)